LungLogic Capstone Project

PPV40V Bench Experiments

PWM–voltage calibration and fake-lung inspiration modelling

Group Meeting

2026-10-06

Two experiments

1. PCB/Arduino PWM → voltage

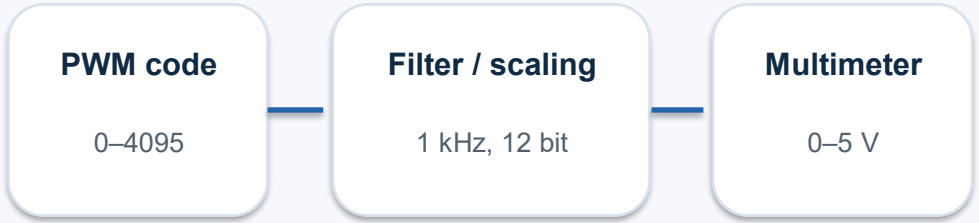
2. PPV voltage → inspiratory flow

Preliminary parameter identification and test-quality review

What do the two experiments establish?

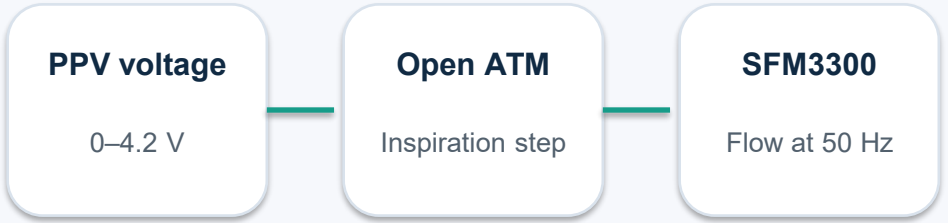
Electrical command conversion and pneumatic system response

Experiment 1: PWM–voltage sweep



- Linearity, steady-state error and sweep repeatability
- Valve position, flow and dynamic delay were not measured

Experiment 2: fake-lung inspiration

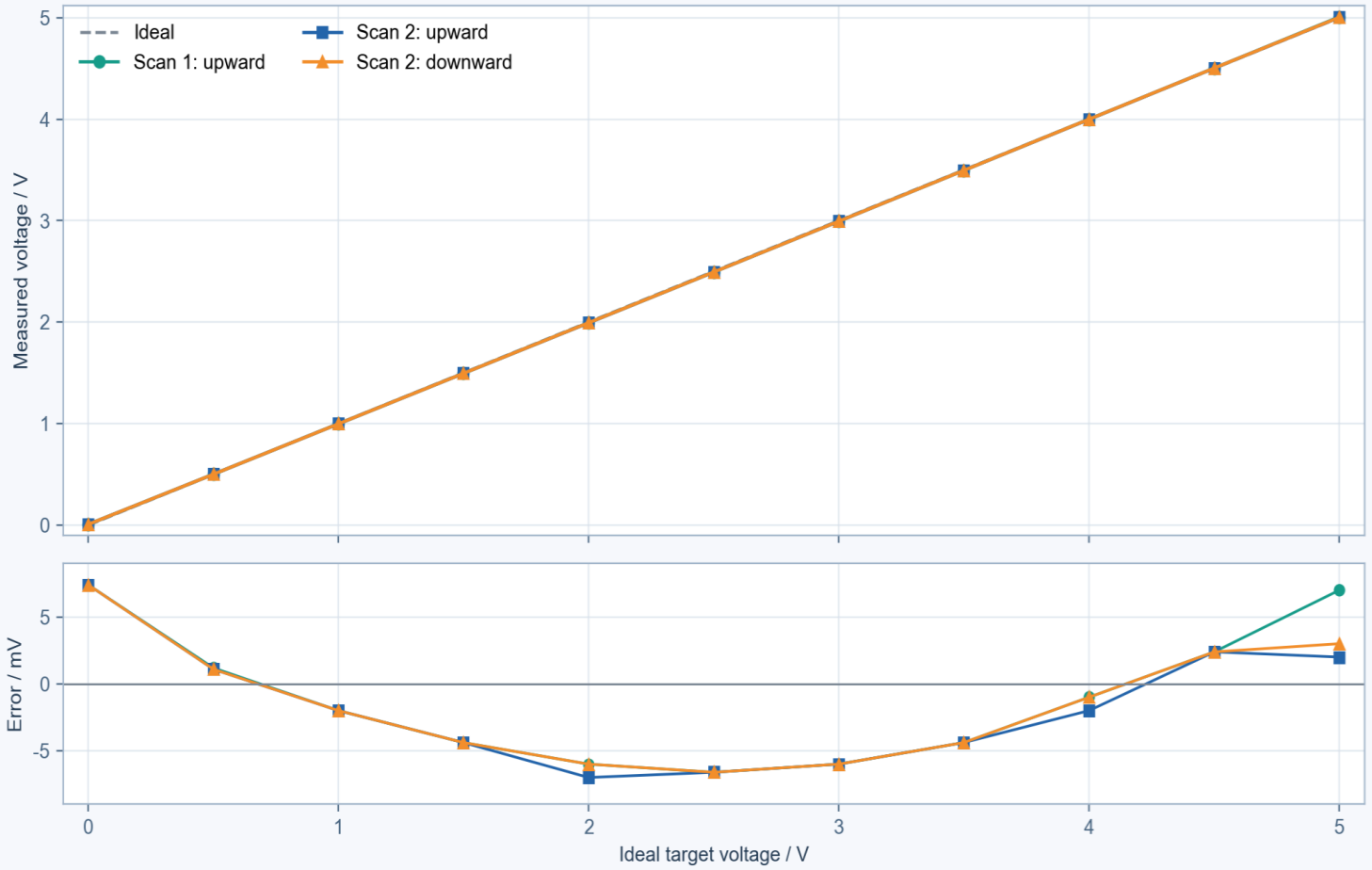


- Time constant, control range, closure voltage and γ
- PPV was preset; its actuation delay cannot be isolated

The electrical calibration supports interpreting the flow plateau as a pneumatic-system effect.

Experiment 1: output voltage closely follows the PWM command

D5: 1 kHz, 12-bit PWM → PCB filtering / scaling → PPV 0–5 V control input



Method

Sweep 0–100%, then 100–0%

Measured results

- Linear R^2: 0.999993
- RMSE: 4.63 mV
- Maximum error: 7.4 mV
- Max. up/down difference: 1.0 mV

Conclusion
Maximum error: 0.148% FS. A linear conversion is adequate for initial modelling.

The small electrical error does not explain the substantial flow nonlinearity in Experiment 2.

Experiment 2: inspiration at a fixed PPV voltage

Manually compress the fake lung, then open ATM to initiate inspiration

1 Compress lung

Manual initial state

2 Close ATM/VAC

Isolate the circuit

3 Set PPV

0–4.2 V

4 Open ATM

Inspiration

5 Record flow

SFM3300, 50 Hz

40 CSV files

Multiple voltages and repeated trials

38 dynamic fits

Exclude 37.csv; retain 42.csv as the closed point

Median R^2: 0.9994

First-order cumulative-volume fits

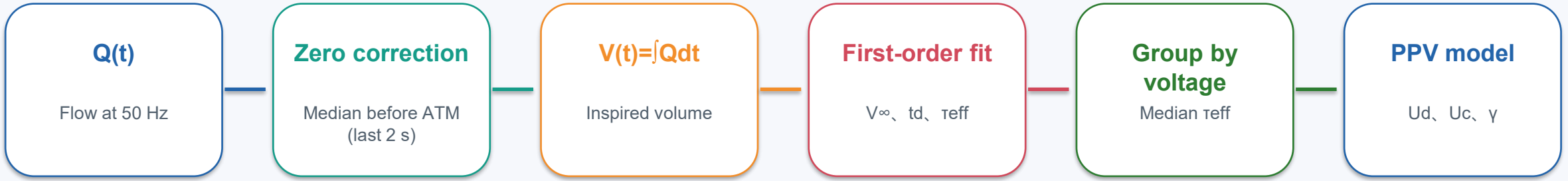
No pressure data

Valve resistance and lung compliance cannot be separated

The fits characterize the combined PPV–tubing–fake-lung dynamics.

a. From flow measurements to initial model parameters

Fit each Q(t) trace, then summarize repeated trials at each voltage using medians



From each trial

- Peak / early flow and cumulative volume V(t)
- τeff, fitted V∞ and combined response delay td

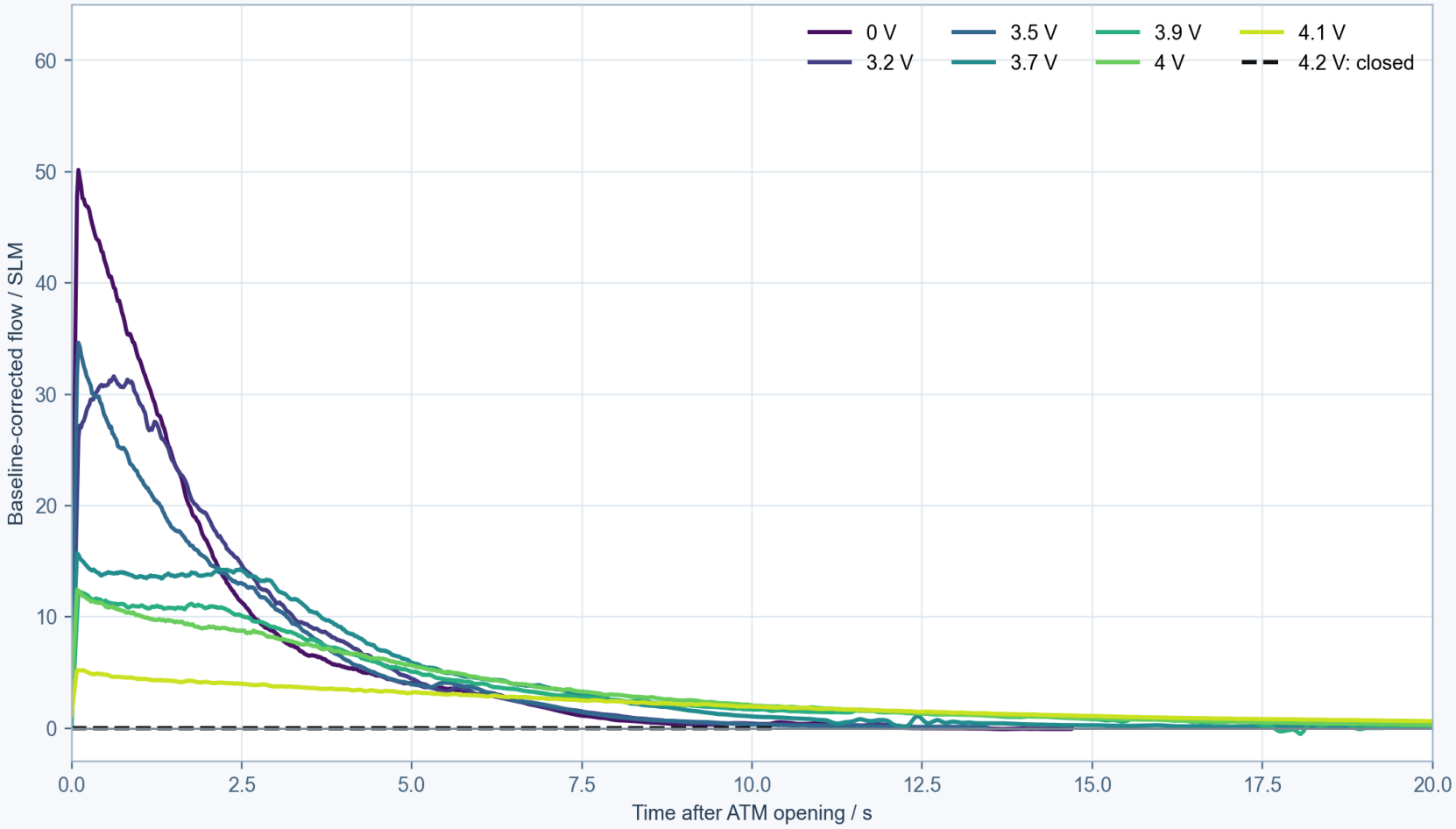
From the voltage sweep

- Fully open baseline time constant τin
- Control onset Ud, closure Uc and opening exponent γ

Peak flow depends strongly on initial compression; γ is therefore estimated mainly from time constants.

Representative flow traces: slower inspiration above 3.3 V

For each voltage, show the trial with a time constant closest to the group median

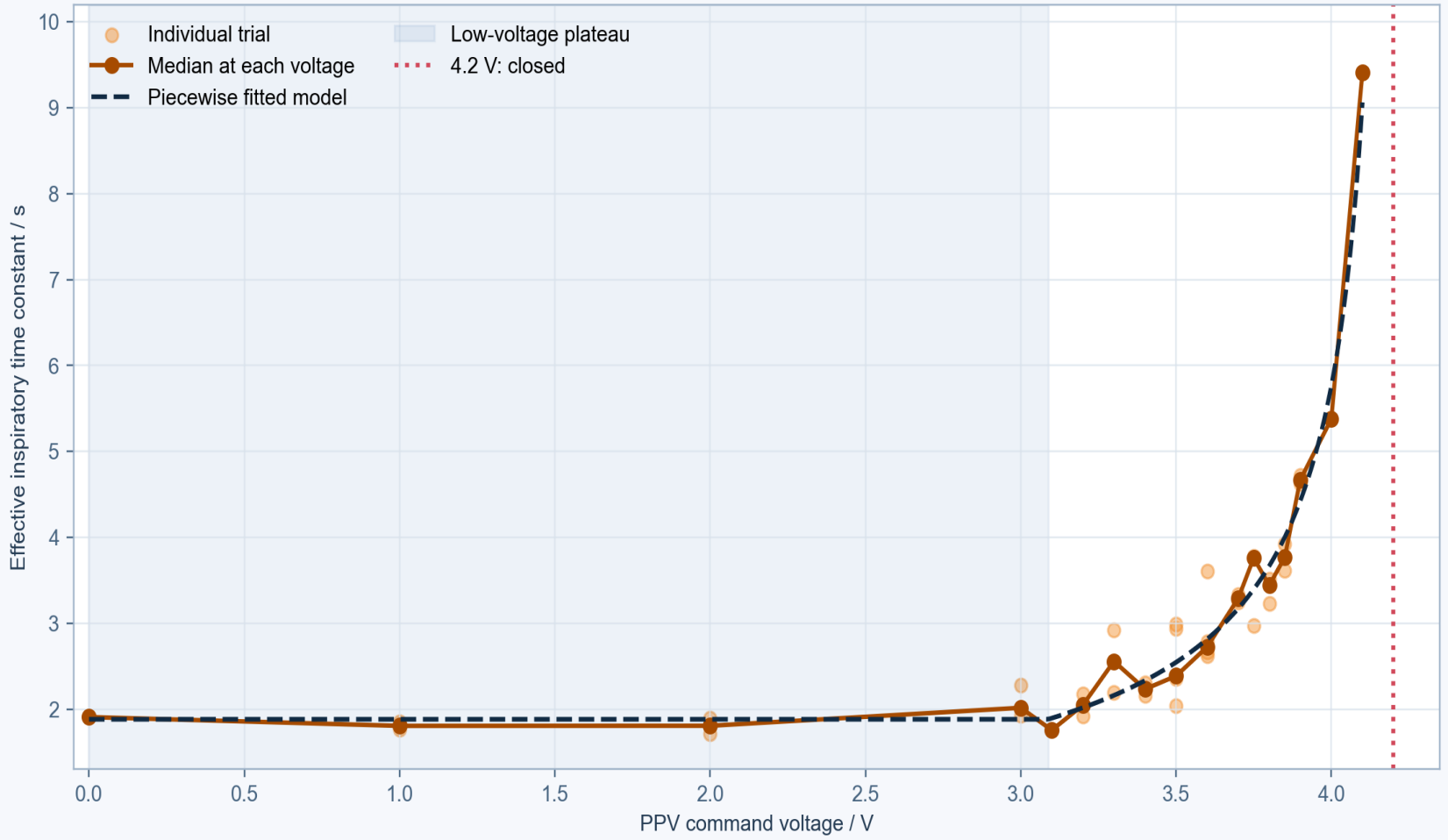


- 0–3.2 V: similar decay times; weak valve restriction
- 3.5–3.9 V: lower peaks and longer inspiration
- 4.1 V: peak ≈5.6 SLM; much slower response
- 4.2 V: flow near baseline noise; closure confirmed
- Initial compression affects peak flow; do not use peaks alone to identify γ

A clear overall trend is present; differences in initial mechanical state likely explain local non-monotonicity.

First-order time-constant identification

Integrate baseline-corrected flow and fit cumulative inspired volume



Fitted model

$$V(t)=V_\infty[1-\exp(-(t-td)/\tau eff)]$$

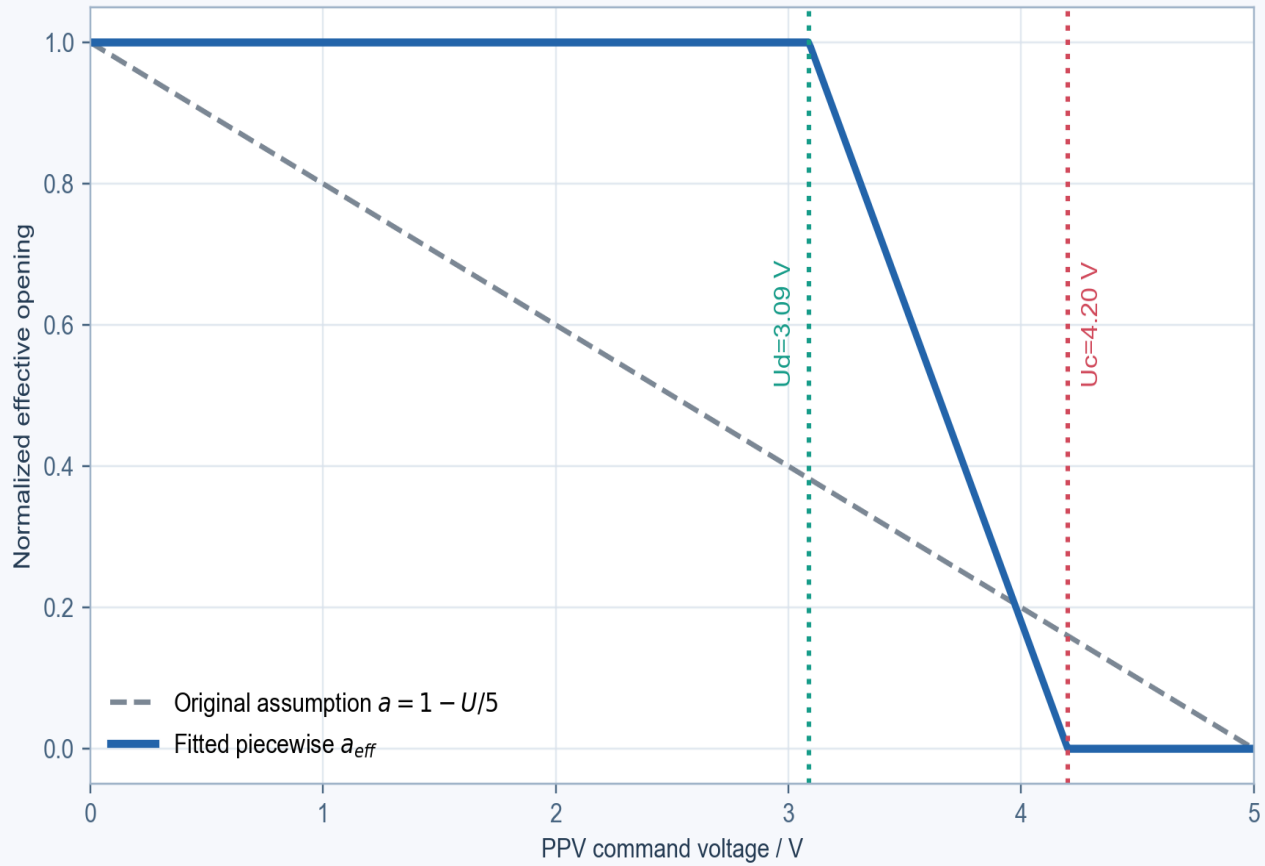
Initial results

- 0–3.2 V: median τ = 1.91 s
- 4.1 V: τ = 9.41 s
- Rapid increase above 3.9 V

Use τin ≈ 1.9 s as the fully open baseline; the time constant rises rapidly above 3.9 V.

PPV nonlinearity: a piecewise effective opening

A low-voltage plateau and closure at 4.2 V make a=1−U/5 unsuitable across the full range



Piecewise model fit

- τin = 1.88 s
- Plateau endpoint Ud = 3.09 V
- Opening exponent γ = 0.653
- Time-constant fit R^2 = 0.986

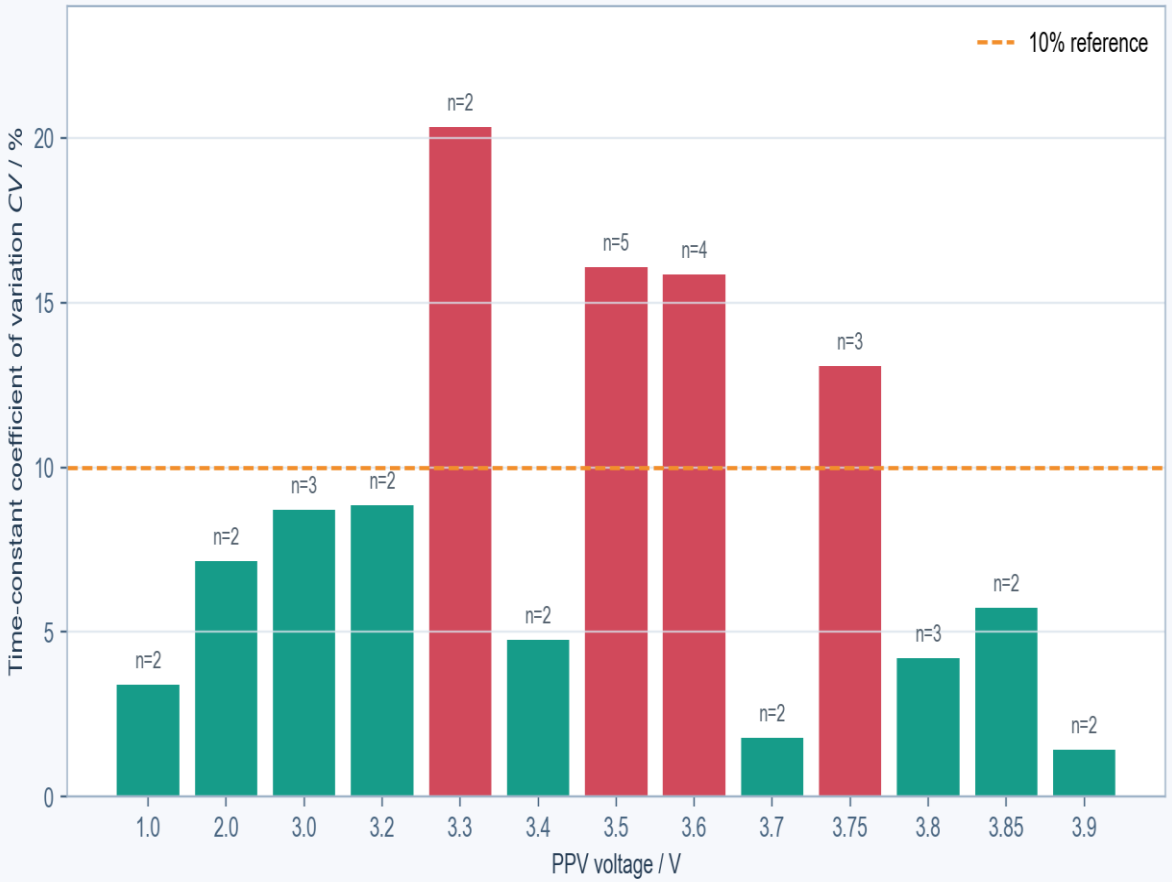
τeff = τin / aeff^γ

Trial-selection sensitivity: γ = 0.60–0.68

Use γ≈0.65 with piecewise aeff; the earlier local γ≈2.3–2.6 is not a global physical parameter.

b. Data screening and improvements

Account for recording errors, confirmed closure and trial-to-trial variability



37.csv

Filename: 3.7 V; recorded PPV: 0 V throughout.

Exclude

42.csv

No flow after repeated ATM_ON; 4.2 V closure confirmed.

Retain closure

33.csv

Differs from 33.1.csv; reverse-flow fluctuations late in the trial.

Low confidence

35.2/35.3.csv

3.5 V set after recording began; lower-flow results.

Low confidence

35.2/35.3 settled for 3.26/3.77 s before ATM; retain with caution.

The main uncertainty is inconsistent initial compression and mechanical state of the fake lung.

c. Limitations: only inspiration was identified

The fitted τ describes the ATM inspiratory path; the vacuum expiratory time constant τout is still unknown

Identified

- Inspiratory flow Q(t) and integrated volume V(t)
- Fully open τin and restricted-opening τeff
- PPV control range, closure voltage and γ

Still unknown

- Expiratory / vacuum-collapse time constant τout
- Isolated PPV command-to-flow delay
- Pressure drop: resistance and compliance cannot be separated

Control implications

- Vacuum is adjustable; τout is less urgent for the current inspiration model
- τout is still needed to assess breathing rate, I:E ratio and cycle consistency
- Identify τout at several vacuum settings

Adjustable vacuum can tune expiration speed; a complete model still needs both inspiratory and expiratory parameters.

Preliminary results: usable initial model parameters

Use these estimates for initial modelling and sensitivity analysis, rather than final calibration

τin ≈ 1.9 s

Fully open; 1.8–2.1 s

Ud ≈ 3.1 V

Control onset

Uc = 4.2 V

Confirmed closure

γ ≈ 0.65

Sensitivity: 0.60–0.70

R^2 = 0.986

τ–voltage fit

Suggested initial inspiration model

$$\tau_{eff} = \tau_{in} / a_{eff}(U)^{\gamma}$$

aeff=1 for U≤3.1 V; decreases linearly to 0 at 4.2 V; closed for U≥4.2 V.

Applications

- Initial feedforward model parameters
- Controller parameter sensitivity tests
- Baseline for subsequent breathing-cycle tests

Apply γ≈0.65 together with the piecewise effective opening aeff(U).

Future work: this afternoon's test plan

Vacuum source received today: measure expiration and evaluate control over consecutive breathing cycles

1. Vacuum / expiration tests

Record flow at several vacuum settings; estimate τout and collapse volume.

2. PPV step response and delay

With ATM open and pressure drop stable, step PPV and record command and flow together.

3. Complete breathing cycles

Run consecutive inspiration / expiration cycles; check phase switching and repeatability.

4. Control performance logging

Log target / measured flow, integrated volume, PPV command and ATM/VAC states together.

Success criteria: estimate τout and demonstrate consistent flow tracking, volume and valve switching over multiple cycles.

Appendix: time constants by voltage

Ranges show the minimum and maximum individual fitted values at each voltage

Voltage / V	n	Median τ / s	τ range / s	Peak flow / SLM	Median volume / mL
0	1	1.91	1.91–1.91	51.7	1602
1	2	1.81	1.76–1.85	47.9	1569
2	2	1.81	1.72–1.90	61.7	1674
3	3	2.02	1.93–2.28	48.5	1569
3.1	1	1.76	1.76–1.76	53.0	1571
3.2	2	2.05	1.92–2.18	41.2	1528
3.3	2	2.56	2.19–2.92	39.7	1539
3.4	2	2.23	2.16–2.31	37.7	1513
3.5	5	2.39	2.04–2.99	35.5	1375
3.6	4	2.72	2.62–3.61	23.0	1396
3.7	2	3.29	3.25–3.33	16.3	1291
3.75	3	3.76	2.97–3.77	17.9	1342
3.8	3	3.44	3.23–3.51	16.5	1323
3.85	2	3.77	3.62–3.92	18.4	1078
3.9	2	4.67	4.62–4.72	12.4	1245
4	1	5.38	5.38–5.38	13.1	1193
4.1	1	9.41	9.41–9.41	5.6	837
4.2	1	Closed	—	≈0	—